
1. Han--Li--Sun $\beta = 0$ Wang Proposition 5.2
“ 2”

2. $\beta > 0$ Han--Li--Sun White
 $\Theta < 2$
 Ψ $\Psi < 1 + \varepsilon_{\mathrm{HLS}}$

1.

M^4 Kähler $M \hookrightarrow \mathbb{R}^N$
 $F : \Sigma \times [0,T) \longrightarrow M$
 $T < \infty$
 $\eta = *F_t^* \omega = \cos \alpha.$

(X_0,T) ψ $\psi \equiv 1$ X_0
$$\rho_{X_0,T}(Y,t) = \frac{1}{4\pi(T-t)} \exp\bigg(-\frac{|Y-X_0|^2}{4(T-t)}\bigg).$$

$$\Theta(X_0,T) := \lim_{t \uparrow T} \int_{\Sigma_t} \psi \, \rho_{X_0,T} \, d\mu_t. \tag{1.1}$$

Huisken $(0,0)$

X_0

Σ_{I} (X_0,T) $x_j \in \Sigma$ $t_j \uparrow T$
 $F(x_j,t_j) \rightarrow X_0,$ $|A|(x_j,t_j) \rightarrow \infty,$

T

$$\Theta < 2$$

$$\begin{aligned} &1 \\ (2.4) \quad &\frac{1}{4\pi} \int_{P_j} e^{-|Y|^2/4} d\mathcal{H}^2(Y) = \frac{1}{4\pi} \int_{\mathbb{R}^2} e^{-|y|^2/4} dy = 1. \end{aligned} \tag{2.5}$$

$$\Theta(X_0,T)=\frac{1}{4\pi}\int e^{-|Y|^2/4}d\|V\|(Y)=\sum_{j=1}^Jm_j. \tag{2.6}$$

$$\begin{aligned} (2.1) \quad &1 \qquad J=1 \quad m_1=1 \\ &\Theta(X_0,T)=1. \end{aligned} \tag{2.7}$$

White

$$\begin{aligned} \text{White} \qquad &\varepsilon_W > 0 \qquad \qquad \qquad 1 + \varepsilon_W \\ (2.7) \quad & \end{aligned}$$

$$\begin{aligned} &\Theta(X_0,T)=1<1+\varepsilon_W, \\ (X_0,T) \qquad (2.2) \qquad &(X_0,T) \end{aligned}$$

$$\begin{aligned} T \qquad &|A| \quad [0,T) \\ \varepsilon > 0 \qquad &[0,T+\varepsilon) \end{aligned}$$

2.2

$$\begin{aligned} \Theta \leq 2 \quad P \qquad &2|P| \qquad \qquad \qquad 2 \\ &\text{“} \qquad \qquad \qquad \text{”} \end{aligned}$$

2.3 Wang Proposition 5.2

$$\begin{aligned} \text{Wang} \qquad &\text{multiplicity one plane} \qquad \qquad \text{White} \qquad \qquad \qquad \text{“} \qquad \qquad \qquad \text{”} \\ &\Theta < 2 \qquad (2.6) \\ \text{“} \qquad \qquad \text{”} \qquad &2 \end{aligned}$$

3. Han--Li--Sun β -

Han--Li--Sun

$$\partial_t F = f = \frac{\cos^2 \alpha \, H - \beta \sin^2 \alpha \, V}{\cos^2 \alpha + \beta \sin^2 \alpha}, \qquad \beta \geq 0, \tag{3.1}$$

$$V=\partial_2\alpha\,v_3+\partial_1\alpha\,v_4 \qquad \beta=0 \quad (3.1) \qquad \partial_t F=H$$

$$\Psi(X_0,t_0;t)=\int_{\Sigma_t}\frac{1}{\cos^p\alpha}\phi(F),\rho_{X_0,t_0}(F,t)\,d\mu_t,\qquad p\geq p_0,\tag{3.2}$$

$$\begin{array}{llll} \bullet \text{ Corollary 4.2} & \varepsilon_{\mathrm{HLS}} > 0 & r > 0 & \\ & \Psi(X_0,t_0;t_0-r^2) < 1 + \varepsilon_{\mathrm{HLS}}, & & \end{array}\tag{3.3}$$

$$(X_0,t_0)\tag{3.1}$$

$$\bullet \text{ Theorem 3.1} \qquad |A| \tag{3.1}$$

$$\mathbf{3.1} \qquad \beta\text{-}$$

$$\begin{array}{llll} F & (3.1) & T & \cos\alpha\geq\delta>0 \\ r_{X_0}>0 & (3.3) & (t_0,r)=(T,r_{X_0}) & T \end{array} \qquad X_0$$

$$\begin{array}{llll} \text{Corollary 4.2} & (X_0,T) & |A| \quad t \uparrow T & M \\ X_0 & & |A| & \text{Theorem 3.1} \end{array}$$

$$\mathbf{4.} \qquad \beta>0 \qquad \mathbf{(3.3)} \qquad \Theta<2$$

$$\begin{array}{llllll} 1. \text{ White} & \beta>0 & (3.1) & H & & \\ 2. \text{ Han--Li--Sun Theorem 5.7} & \lambda\text{-} & & & \text{K\"ahler} & \\ 3. & \text{K\"ahler} & c=\cos\alpha_P\in(0,1] & 1 & (3.2) & c^{-p} \end{array}$$

$$\Theta<2$$

$$\text{varifold}$$

$$\Psi<1+\varepsilon_{\mathrm{HLS}}.$$

$$4.$$

$$\text{Wang Proposition 5.2} \qquad \beta>0$$

$$\Theta<2$$

- $\cos \alpha_P = 1$
- (3.1) “ ”
- (3.3) $1 + \varepsilon_{\text{HLS}}$

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5.

$\beta = 0$ MCF	White	$\Theta < 2$ Wang	$\Theta \leq 2$ 1
$\beta > 0$ Han--Li--Sun	Han--Li--Sun Corollary 4.2	$\Psi < 1 + \varepsilon_{\text{HLS}}$	Theorem 3.1
$\beta > 0$ $\Theta < 2$	White	1	

6.

2.1			
$F_t : \Sigma \rightarrow M^4$	Kähler	T	$*F_t^* \omega \geq \delta > 0$
$ A_t ^2 \leq C/(T - t)$	(X_0, T)	$\Theta(X_0, T) < 2$	$\varepsilon > 0$
$[0, T + \varepsilon)$			
$\beta > 0$	3.1	$\Psi < 1 + \varepsilon_{\text{HLS}}$	4
$\Theta < 2$			

[1] M.-T. Wang, *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds*, J. Differential Geom. 57 (2001), 301--338 Proposition 5.2

[2] X. Han, J. Li and J. Sun, *Gradient Flow for β -Symplectic Critical Surfaces*, Ann. Inst. H. Poincaré C Anal. Non Linéaire 41 (2024), 1083--1116 Theorems 3.1, 4.1, 5.7 Corollary 4.2

- [3] B. White, *A Local Regularity Theorem for Mean Curvature Flow*, Ann. of Math. 161 (2005), 1487--1519